

CSE2315 Lab Course

Solutions 6

deadline: March 20, 2026, 8:45

1. Suppose we have the following language:

$$E_{\text{GNFA}} = \{\langle G \rangle \mid G \text{ is a GNFA such that } L(G) = \emptyset\}.$$

Show that E_{GNFA} is decidable. You are allowed to make use of the deciders implied by the text in the book (e.g., Section 4.1).

Solution: The following TM M decides E_{GNFA} :

$M =$ "On input $\langle G \rangle$:

1. Convert G into a DFA D .
2. Run the decider for E_{DFA} on $\langle D \rangle$.
3. If the decider accepts, accept. Otherwise, reject."

2. Consider the language

$$L = \{\langle M, w \rangle \mid M \text{ halts on } w \text{ and moves the head at most } |w|^2 \text{ steps}\}.$$

Is this language decidable? Explain.

Solution: L is decidable. Remember that a TM must move the head every step. Thus we can simulate M on w for at most $|w|^2$ steps. If it halts in that time, we accept. Else M either has looped, or will do more than $|w|^2$ steps with the head in the next step. In both of these cases, we can reject with certainty.

3. *Old exam question, endterm 20-21:* Show the claim is true or give a counterexample and explanation to show the claim is false, starting your answer with either *True* or *False*.

If there is a Turing reduction from L to E_{CFG} , then $\bar{L}$ is Turing-recognisable.

Solution: True. Since E_{CFG} is decidable and L is reducible to E_{CFG} and decidability is downwards closed, it follows that L is decidable and thus L is both Turing and co-Turing-recognisable, and thus $\bar{L}$ is Turing-recognisable.

4. *Old exam question, endterm 22-23:*

Consider the following language:

$$L = \{\langle M, q \rangle \mid M \text{ is a Turing Machine and } q \text{ is a useless state in } M\}$$

A useless state in a Turing Machine is one that is never entered on any input string.

Prove L is undecidable, using a Turing reduction $E_{\text{TM}} \leq_T L$.

Solution: We can use a reduction from E_{TM} :

Consider for the sake of contradiction that L is decidable and let R be a Turing machine that decides it. Now we construct a decider S for E_{TM} :

$S =$ "On input $\langle M \rangle$:

1. Run R on $\langle M, q \rangle$, where q is M 's accept state.
2. If R accepts, accept. Otherwise, reject.

If $L(M) = \emptyset$, R will accept $\langle M, q \rangle$ as there do not exist words for which m reaches the accept state. If $L(M) \neq \emptyset$, R will reject there exists a word for which M reaches the accept state.

Thus S decides E_{TM} . But E_{TM} is undecidable. Contradiction! Hence no such machine R can exist and thus L must be undecidable.

Taken from: Sipser exercise 5.29

5. Old exam question, resit 17-18: Suppose we have the following language

$$L = \{\langle M, v, G \rangle \mid M \text{ outputs } G \text{ on the tape, which is a context free grammar that accepts } v, \text{ when run on input } v.\}$$

We intend to prove that this language is undecidable. To that end we use a mapping reduction from $HALT_{TM}$ to L . The reduction f is computed by the following TM F :

$F =$ "On input $\langle M, w \rangle$:

1. Construct the following TM M' :

$M' =$ "On input x :

1. Clear the input from the tape.
2. Run M on w .
3. See question a.
4. See question a.

2. Write $\langle M', a, G \rangle$ to the output tape, where G is a grammar with one rule: $S \rightarrow a$ and starting variable S ."

- (a) What should be done in steps 3 and 4 of M' ?

Solution:

3. Create a grammar G' with starting variable S and one rule: $S \rightarrow a$.
4. Write the grammar G' to the tape and accept."

- (b) Provide a proof showing f satisfies the requirements of a mapping reduction.

Solution:

Proof. We need to show that f is computable and reliable. F computes f , so f is computable. f is also reliable, because:

- $\langle M, w \rangle \in HALT_{TM} \Rightarrow f(\langle M, w \rangle) \in L$ holds, since if M halts on w , M' leaves G on the tape on input a and this is exactly what it is supposed to leave. So, $\langle M, a, G \rangle \in L$.
- $\langle M, w \rangle \notin HALT_{TM} \Rightarrow f(\langle M, w \rangle) \notin L$ holds, since if M does not halt on w , M' will never write the correct output to the tape.

□

6. Old exam question, resit 20-21:

- (a) Let MIA_{TM} be the language consisting of 3 TM's and a word such that the word is recognized by at least 2 of the 3 TM's, formally described as $MIA_{TM} = \{\langle M_1, M_2, M_3, w \rangle \mid M_1, M_2 \text{ and } M_3 \text{ are TM's and } w \text{ is a word such that } w \in (L(M_1) \cap L(M_2)) \cup (L(M_1) \cap L(M_3)) \cup (L(M_2) \cap L(M_3))\}$. Show that this is a recognizable language.

Solution: We run M_1 , M_2 and M_3 on w at the same time using dovetailing (meaning we let the machines all take one step in turn). As soon as 2 of the machines have accepted, we accept.

- (b) Show that MIA_{TM} is undecidable by constructing a Turing reduction from A_{TM} .

Solution: Assume, for the sake of contradiction, that MIA is decidable. We now use it to construct a machine that decides A_{TM} . Given that MIA is decidable, let R be a decider for MIA.

TM N , on input $\langle M, w \rangle$:

- Construct a TM M_1 such that $L(M_1) = \emptyset$ and a TM M_2 such that $L(M_2) = \Sigma^*$
- Run R on $\langle M, M_1, M_2, w \rangle$
- If R accepts, accept. Else, reject.

N decides A_{TM} , but A_{TM} is undecidable. This is a contradiction, thus our assumption that MIA_{TM} is decidable must have been incorrect. So, by contradiction, we have shown that MIA is undecidable.

- (c) Show that MIA_{TM} is not co-turing recognizable.

Solution: We know MIA_{TM} is turing recognizable and not decidable, from which follows immediately that MIA_{TM} is not co-turing recognizable

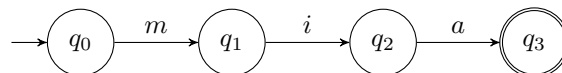
7. Revision, old exam question, midterm 19-20:

- (a) Assume that we define an expanded NFA model by imposing the additional requirement that $|F| = 1$, and we call this an SNFA. Are the SNFA and NFA models equally expressive? Start your answer with "Yes" or "No". If you answer yes, explain how an arbitrary SNFA can be rewritten as an NFA and vice versa. If you answer no, give an example of a language that can be accepted by an NFA, but not by an SNFA or vice versa.

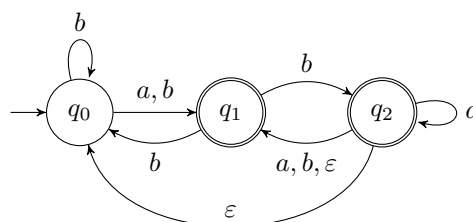
Solution: Yes. Every SNFA is trivially an NFA without making any modifications, it is simply an NFA where $|F| = 1$ happens to hold. To go from an NFA to an SNFA make an extra state q_a and add an epsilon transition from every $q \in F$. Now for the NFA $N = (Q, \Sigma, \delta, q_0, F)$ the SNFA $S = (Q, \Sigma, \delta \cup \delta_2, q_0, \{q_a\})$ is an SNFA such that $L(N) = L(S)$.

- (b) Give an NFA of at most 4 states that accepts the word *mia* and rejects the word *may*. No other restrictions on the language apply. A transition diagram suffices.

Solution:



- (c) Consider the NFA N depicted as:



Using the method from Sipser and leaving out unreachable states, construct a new DFA D , such that $L(D) = L(N)$. A transition diagram and a short explanation suffice.

